Dynamic Data Masking

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Section Introduction-Dynamic Data Masking

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What is Dynamic Data Masking?

Snowflake Dynamic Data masking is a column level security feature that uses masking policies to selectively mask plain text data in tables or views at query time.

FIRST_NAME	LAST NAME	EMAIL	SSN	DATE_OF_BIRTH
STEVEN	KING	SKING	*********	9999-01-01
NEENA	KOCHHAR	SKOCHHAR	*********	9999-01-01
LEX	DE HAAN	LDEHAAN	*********	9999-01-01
ALEXANDER	HUNOLD	AHUNOLD	*********	9999-01-01
BRUCE	ERNST	BERNST	*********	9999-01-01
DAVID	AUSTIN	DAUSTIN	*********	9999-01-01
DIANA	LORENTZ	DLORENTZ	*********	9999-01-01
NANCY	GREENBERG	NGREENBERG	*********	9999-01-01
DANIEL	FAVIET	DFAVIET	*********	9999-01-01
JOHN	CHEN	JCHEN	*********	9999-01-01

What is Dynamic Data Masking?

Types of data that are masked are

- PII (Personally Identifiable information)
- PCI (Payment Card Industry)

PII (Personally Identifiable information)

It is any information that can be used to identify an individual either directly or indirectly.

Example of PII data include

- Full name
- Social Security number
- Driver's license number
- Passport number
- Phone number
- Date of birth

What is Dynamic Data Masking?

PCI (Payment Card Industry)

Refers to the global standard for the security of credit card transactions.
It outlines the storage requirement of credit card data when stored in databases and Data Masking can protect the data.

Example of PCI data include

- Credit Card Number
- Card Holder Name
- Card Expiration Date
- Cardholder Personal Identification Number(PIN)

Summary

- Snowflake Dynamic Data Masking is a column level security feature that uses masking policies to selectively mask plain text data in tables or views at query time.
- PII (Personally Identifiable Information) and PCI(Payment Card Industry) data are generally masked to protect the data from unauthorized access.
- Examples of PII data include
  - Full name
  - Social Security number
  - Driver's license number
  - Passport number
  - Phone number
  - Date of birth
- Examples of PCI data Include
  - Credit Card Number
  - Card Holder Name
  - Card Expiration Date
  - Cardholder Personal Identification Number(PIN)
- Masking PII and PCI data help prevent identity theft and financial fraud .
- Masking policies when created and SET on a column in a table mask the data differently depending on the role being assumed.

Summary

- Masking policies when created must have the same data type between the input value, return value and the data type of the column being masked.
- When masking columns the masking values used to replace the column data must be of the same data types as the column being masked , so if the column being masked is DATE then the replacement masked values must also be DATE.
- A given Masking functions can be used on 1 or more columns in the same table or a different table.
- UDF(User Defined Functions) can be created and used inside a data masking policy. This is generally done when the data masking logic is complex and moving the logic to a UDF makes code management easier.
- SHA2 is a hashing algorithm which can be used to HASH data. SHA2 always returns the same Hash value output for the same input to the function.
- Masking functions can be UNSET from columns if you don't need them to mask data anymore.
- A table with masking policies set on it can be dropped. Dropping the table does not cause the masking policies to be dropped.
- Data returned by the WHERE clause in the SELECT statement will see the masked data and the evaluation will be based on the masked data

Summary

- Users and roles used in ETL batch jobs should be able to see the unmasked data so that it can evaluate and process the data based on the unmasked data else the outcome of the ETL batch job may not be accurate.
- A new table created from another table that has masked data will have the data as masked. The Masking policies are not inherited by the new table and the masked data is permanently changed.
- Just like tables views can be masked too, masking views instead of tables and granting access to roles allows for more granular control of the data.
- Snowflake permits conditional data masking in which you can mask a column based on the data values of a different column. The columns used in the masking and the condition column must be provided as input parameter to the masking policy.
- The SET statement should refer to both the parameters used in the conditional masking policy when setting the masking policy.

Thank you

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